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Inventor(s)	JUNG; Minsu et al.

ELECTRONIC DEVICE COMPRISING WATERPROOF STRUCTURE

Abstract

An electronic device including: a first housing including at least one first electronic component, the first housing having a through-hole therein; a second housing including at least one second electronic component; a hinge device that rotatably connects the first housing to the second housing; a flexible printed circuit board that penetrates through the through-hole in the first housing, and electrically connects the at least one first electronic component and the at least one second electronic component; a stiffener on the flexible printed circuit board; and a waterproof structure in the through-hole, wherein the waterproof structure includes: a first waterproof member coupled with the stiffener to form a loop structure having an opening, and a second waterproof member on the first waterproof member and covering the opening, and wherein the stiffener and the first waterproof member surround a periphery of the through-hole.

Inventors:	**JUNG; Minsu (Suwon-si, KR), KIM; Gyeongtae (Suwon-si, KR)**
Applicant:	**SAMSUNG ELECTRONICS CO., LTD.** (Suwon-si, KR)
Family ID:	**1000008627786**
Assignee:	**SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)**
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a by-pass continuation application of International Application No. PCT/KR2023/017423, filed on Nov. 2, 2023, which is based on and claims priority to Korean Patent Application No. 10-2022-0164311, filed on Nov. 30, 2022, and Korean Patent Application No. 10-2022-0181512, filed on Dec. 22, 2022, in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein their entireties.

BACKGROUND
1. Field
[0002] The disclosure relates to an electronic device including a waterproof structure.
2. Description of Related Art
[0003] As the functional gap among electronic devices from different manufacturers has significantly decreased, electronic devices are gradually becoming slimmer to satisfy consumer purchasing desires. At the same time, rigidity of electronic devices is increasing, and their design aspects are being strengthened, while improvements are being made to differentiate their functional elements. An electronic device may have a transformable structure that allows for convenient portability and the ability to use a large-screen display. For example, an electronic device may be a foldable electronic device where a first housing is rotatably connected to a second housing through a hinge device (e.g., a hinge module, a hinge structure, or a hinge assembly). The foldable electronic device needs to ensure efficient layout of internal electronic components and to secure operational reliability in response to folding operations.
[0004] The first housing and the second housing are rotatably coupled through the hinge device, thereby ensuring operational reliability for the folding state and/or unfolding state. The foldable electronic device may be operated in an in-folding manner and/or an out-folding manner by rotating the first housing relative to the second housing through the hinge device in a range from 0 to 360 degrees. The foldable electronic device may include a flexible display that is disposed to be supported by the first housing and the second housing in a state where it is opened at 180 degrees.
[0005] The foldable electronic device may include a flexible printed circuit board (FPCB) that electrically connects the first housing and the second housing. The flexible printed circuit board may be disposed in the inner space of a hinge housing that accommodates the hinge device, and include a bending portion formed in a bent shape (e.g., a shape bent at least once) that is at least partially elastic (e.g., resiliency) to accommodate a length that is deformed according to the folding operation of the foldable electronic device. The flexible printed circuit board may be disposed such that one end penetrates through at least a part of the first housing and then is electrically connected to a first printed circuit board (e.g., the main printed circuit board), and the other end penetrates through at least a part of the second housing and then is electrically connected to a second printed circuit board (e.g., the sub printed circuit board).
[0006] The foldable electronic device may include a fixation structure formed in a portion that is introduced into the first housing and second housing from the bending portion. For example, the

fixation structure may be configured in a manner where a stiffener disposed on the flexible printed circuit board is attached to at least a part of the first housing or second housing. The foldable electronic device may include a waterproof structure to prevent or reduce the ingress of external moisture or foreign substances, which is provided by the flexible printed circuit board disposed in a manner that penetrates through each housing. Such a waterproof structure may be configured in a manner that seals a through-hole through which the flexible printed circuit board penetrates the first housing and second housing.

[0007] However, with the slimming down of the foldable electronic device or the diversification of the folding method of the flexible display (e.g., folding into a droplet shape), a space in which the bending portion of the flexible printed circuit board is accommodated, from the fixation structure to the hinge housing inner space (e.g., space where elastic deformation of a bending area is required), may become narrower. For example, as a space of a folding area of the flexible display increases, the space required for the bending portion of the flexible printed circuit board may also increase, which may make it difficult to slim down the electronic device.

SUMMARY

[0008] Provided is an electronic device including a waterproof structure that may assist in slimming down the electronic device.

[0009] Further, provided is an electronic device including a waterproof structure that may assist in slimming down the electronic device, while simultaneously inducing smooth operation of the electronic device by securing a space for the elastic deformation of the flexible printed circuit board.

[0010] Further, provided is an electronic device including a waterproof structure that may assist in improving assembly.

[0011] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

[0012] According to an aspect of the disclosure, an electronic device including: a first housing including at least one first electronic component, the first housing having a through-hole therein; a second housing including at least one second electronic component; a hinge device that rotatably connects the first housing to the second housing; a flexible printed circuit board that penetrates through the through-hole in the first housing, and electrically connects the at least one first electronic component and the at least one second electronic component; a stiffener on the flexible printed circuit board; and a waterproof structure in the through-hole, wherein the waterproof structure includes: a first waterproof member coupled with the stiffener to form a loop structure having an opening, and a second waterproof member on the first waterproof member and covering the opening, and wherein the stiffener and the first waterproof member surround a periphery of the through-hole.

[0013] The first waterproof member may have an opened side that is closed by coupling

[0014] with the stiffener.

[0015] The opening may overlap the through-hole.

[0016] The opening may be smaller than the through-hole.

[0017] The first waterproof member may include: a body including a rigid material; and a first coupling protrusion and a second coupling protrusion respectively coupled to opposing sides of the body and each including an elastic material.

[0018] The stiffener may include a first coupling groove and a second coupling groove respectively at opposite ends of the stiffener, the first coupling groove being coupled to the first coupling protrusion and the second coupling groove being coupled to the second coupling protrusion.

[0019] The rigid material of the body may include a metal, and the elastic material of the first and second coupling protrusions may include at least one of rubber, urethane, or silicone.

[0020] The electronic device may further include a third waterproof member between the stiffener and the first waterproof member, wherein the third waterproof member covers the through-hole or

the opening.
[0021] At least one of the second waterproof member or the third waterproof member may include a waterproof tape.
[0022] The electronic may further include a fixing bracket above the second waterproof member, overlapping at least a portion of the second waterproof member, and fixed to the first housing.
[0023] The fixing bracket may include: a plate portion that overlaps the second waterproof member, and a pair of extensions extending from opposite ends of the plate portion, and wherein the pair of extensions may be fixed to the first housing.
[0024] The fixing bracket may be fixed in such a manner that the pair of extensions are fastened to the first housing through screws.
[0025] The through-hole may be in a recess in the first housing.
[0026] The waterproof structure may not protrude from the recess.
[0027] The at least one first electronic component may include a main printed circuit
[0028] board, and wherein the at least one second electronic component may include a sub printed circuit board.
[0029] According to an aspect of the disclosure, an electronic device includes: a first housing including at least one first electronic component, the first housing having a first through-hole therein; a second housing including at least one second electronic component, the second housing having a second through-hole therein; a hinge device that rotatably connects the first housing to the second housing, a flexible printed circuit board that penetrates through the first through-hole in the first housing and the second through-hole in the second housing, and electrically connects the at least one first electronic component and the at least one second electronic component; a stiffener on the flexible printed circuit board; a first waterproof structure in the first through-hole; and a second waterproof structure in the second through-hole, wherein each of the first waterproof structure and the second waterproof structure includes: a first waterproof member coupled with the stiffener to form a loop structure having an opening, a second waterproof member on the first waterproof member and covering the opening, a third waterproof member between the stiffener and the first waterproof member, and a fixing bracket overlapping at least a portion of the second waterproof member and fixed to a corresponding one of the first housing or the second housing, wherein the stiffener and the first waterproof member surrounds a first periphery of the first through-hole and a second periphery of the second through-hole.
[0030] The first waterproof member may have an opened side that is closed by coupling with the stiffener.
[0031] The opening may overlap at least one of the first through-hole or the second through-hole.
[0032] The first waterproof member may include: a body including a rigid material, and a first coupling protrusion and a second coupling protrusion respectively coupled to opposing sides of the body and each including an elastic material.
[0033] The stiffener may include a first coupling groove and a second coupling groove respectively at opposite ends of the stiffener, the first coupling groove being coupled to the first coupling protrusion and the second coupling groove being coupled to the second coupling protrusion.
[0034] The electronic device according to one or more example embodiments of the present disclosure, where the fixation structure and waterproof structure for the flexible printed circuit board are implemented together at the through-hole, may improve flexibility by securing the driving length of the flexible printed circuit board according to the folding operation of the electronic device, and may assist in slimming down the electronic device without increasing its thickness. In addition, the waterproof structure of the flexible printed circuit board is disposed together with the fixation structure, which may assist in improving assembly.
[0035] In addition, various effects that may be directly or indirectly identified through the present disclosure may be provided.
[0036] The effects obtained by the present disclosure are not limited to the aforementioned effects,

and other effects, which are not mentioned above, will be understood by those skilled in the art from the following description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0037] The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0038] FIG. **1**A is a perspective view of a front surface of an electronic device illustrating an unfolding state (or flat state) according to one or more embodiments of the present disclosure;

[0039] FIG. **1**B is a top view illustrating the front surface of the electronic device in the unfolding state according to one or more embodiments of the present disclosure;

[0040] FIG. **1**C is a top view illustrating a rear surface of the electronic device in the unfolding state according to one or more embodiments of the present disclosure;

[0041] FIG. **2**A is a perspective view of the electronic device in a folding state according to one or more embodiments of the present disclosure;

[0042] FIG. **2**B is a perspective view of the electronic device in an intermediate state according to one or more embodiments of the present disclosure;

[0043] FIG. **3** is an exploded perspective view of the electronic device according to one or more embodiments of the present disclosure;

[0044] FIG. **4** is a partial cross-sectional view of the electronic device taken along line **4-4** of FIG. **1**C according to one or more embodiments of the present disclosure;

[0045] FIG. **5**A is an exploded perspective view of a waterproof structure according to one or more embodiments of the present disclosure;

[0046] FIG. **5**B illustrates the assembly sequence of the waterproof structure according to one or more embodiments of the present disclosure; and

[0047] FIGS. **6**A to **6**G illustrate the coupling process according to one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

[0048] FIG. **1**A is a perspective view of an electronic device illustrating an unfolded state (or flat state) of the electronic device according to various embodiments of the disclosure. FIG. **1**B is a plan view illustrating a front surface of an electronic device in an unfolded state of the electronic device according to various embodiments of the disclosure. FIG. **1**C is a plan view illustrating a rear surface of an electronic device in an unfolded state according to various embodiments of the disclosure.

[0049] FIG. **2**A is a perspective view of an electronic device illustrating a folded state of the electronic device according to various embodiments of the disclosure. FIG. **2**B is a perspective view of an electronic device illustrating an intermediate state of the electronic device according to various embodiments of the disclosure.

[0050] With reference to FIGS. **1**A to **2**B, an electronic device **100** may include first and second housings **110** and **120** (e.g., foldable housing structure) combined with each other in a foldable manner based on a hinge device (e.g., hinge device **140** of FIG. **1**B). In an embodiment, the hinge device (e.g., hinge device **140** of FIG. **1**B) may be disposed in an X-axis direction or in a Y-axis direction. According to an embodiment, the electronic device **100** may include a first display **130** (e.g., flexible display, foldable display, or main display) disposed in an area (e.g., recess) formed by the first and second housings **110** and **120**. According to an embodiment, the first housing **110** and the second housing **120** may be disposed on both sides around a folding axis F, and may have a substantially symmetrical shape about the folding axis F. According to an embodiment, an angle or a distance between the first housing **110** and the second housing **120** may differ depending on the state of the electronic device **100**. For example, depending on whether the electronic device is in an unfolded state (or flat state), in a folded state, or in an intermediate state, the angle or the distance between the first housing **110** and the second housing **120** may differ.

[0051] According to an embodiment, in the unfolded state of the electronic device **100**, the first housing **110** may include a first surface **111** directed in a first direction (e.g., front direction) (z-axis direction), and a second surface **112** directed in a second direction (e.g., rear direction) (−z-axis direction) opposite to the first surface **111**. According to an embodiment, in the unfolded state of the electronic device **100**, the second housing **120** may include a third surface **121** directed in the first direction (z-axis direction), and a fourth surface **122** directed in the second direction (−z-axis direction). According to an embodiment, in the unfolded state of the electronic device **100**, the first surface **111** of the first housing **110** and the third surface **121** of the second housing **120** may be directed in substantially the same first direction (z-axis direction). In an embodiment, in the folded state of the electronic device **100**, the first surface **111** of the first housing **110** and the third surface **121** of the second housing **120** may face each other. According to an embodiment, in the unfolded state of the electronic device **100**, the second surface **112** of the first housing **110** and the fourth surface **122** of the second housing **120** may be directed in substantially the same second direction (−z-axis direction). In an embodiment, in the folded state of the electronic device **100**, the second surface **112** of the first housing and the fourth surface **122** of the second housing **120** may be directed in opposite directions. For example, in the folded state of the electronic device **100**, the second surface **112** may be directed in the first direction (z-axis direction), and the fourth surface **122** may be directed in the second direction (−z-axis direction). In this case, the first display **130** may not be viewed from outside (in-folding manner). In an embodiment, the electronic device **100** may be folded so that the second surface **112** of the first housing **110** and the fourth surface **122** of the second housing **120** face each other. In this case, the first display **130** may be disposed to be viewed from the outside (out-folding manner).

[0052] According to various embodiments, the first housing **110** (e.g., first housing structure) may include a first lateral member **113** forming the appearance of the electronic device **100**, and a first rear cover **114** combined with the first lateral member **113**, and forming at least a part of the second surface **112** of the electronic device **100**. According to an embodiment, the first lateral member **113** may include a first side surface **113***a*, a second side surface **113***b* extending from one end of the first side surface **113***a*, and a third side surface **113***c* extending from the other end of the first side surface **113***a*. According to an embodiment, the first lateral member **113** may be formed in a quadrangular (e.g., square or rectangular) shape through the first side surface **113***a*, the second side surface **113***b*, and the third side surface **113***c*.

[0053] According to various embodiments, the second housing **120** (e.g., second housing structure) may include a second lateral member **123** forming the appearance of the electronic device **100** at least partly, and a second lateral cover **124** combined with the second lateral member **123**, and forming at least a part of the fourth surface **122** of the electronic device **100**. According to an embodiment, the second lateral member **123** may include a fourth side surface **123***a*, a fifth side surface **123***b* extending from one end of the fourth side surface **123***a*, and a sixth side surface **123***c* extending from the other end of the fourth side surface **123***a*. According to an embodiment, the second lateral member **123** may be formed in a quadrangular shape through the fourth side surface **123***a*, the fifth side surface **123***b*, and the sixth side surface **123***c*.

[0054] According to various embodiments, the first and second housings **110** and **120** are not limited to the illustrated shapes and combinations, but may be implemented by combinations and/or compositions of other shapes or components. In an embodiment, the first lateral member **113** may be integrally formed with the first rear cover **114**, and the second lateral member **123** may be integrally formed with the second rear cover **124**.

[0055] According to various embodiments, in the unfolded state of the electronic device **100**, the second side surface **113**_b_ of the first lateral member **113** and the fifth side surface **123**_b_ of the second lateral member **123** may be connected to each other without a gap. According to an embodiment, in the unfolded state of the electronic device **100**, the third side surface **113**_c_ of the first lateral member **113** and the sixth side surface **123**_c_ of the second lateral member **123** may be connected to each other without a gap. According to an embodiment, in the unfolded state of the electronic device **100**, the sum of the lengths of the second side surface **113**_b_ and the fifth side surface **123**_b_ may be configured to be longer than the length of the first side surface **113**_a_ and/or the fourth side surface **123**_a_. According to an embodiment, in the unfolded state of the electronic device **100**, the sum of the lengths of the third side surface **113**_c_ and the sixth side surface **123**_c_ may be configured to be longer than the length of the first side surface **113**_a_ and/or the fourth side surface **123**_a_.

[0056] With reference to FIGS. **2**A and **2**B, the first lateral member **113** and/or the second lateral member **123** may be formed of metal, or may further include polymer that is injected into metal. According to an embodiment, the first lateral member **113** and/or the second lateral member **123** may include at least one conductive part **116** and/or **126** electrically segmented through at least one segment part **1161**, **1162** and/or **1261**, **1262** formed of polymer. In this case, the at least one conductive part **116** and/or **126** may be electrically connected to a wireless communication circuit included in the electronic device **100**, and thus may be used as at least a part of an antenna that operates in at least one designated band (e.g., legacy band).

[0057] According to various embodiments, the first rear cover **114** and/or the second rear cover **124** may be formed of, for example, at least one of coated or colored glass, ceramic, polymer, or metal (e.g., aluminum, stainless steel (STS), or magnesium) or a combination of at least two thereof.

[0058] According to various embodiments, the first display **130** may be disposed to extend from the first surface **111** of the first housing **110** to at least a part of the third surface **121** of the second housing **120** across the hinge device (e.g., hinge device **140** of FIG. **1**B). In an embodiment, the first display **130** may include a first area **130**_a_ substantially corresponding to the first surface **111**, a second area **130**_b_ corresponding to the second surface **121**, and a third area **130**_c_ (e.g., flexible area or folding area) connecting the first area **130**_a_ and the second area **130**_b_ to each other. According to an embodiment, the third area **130**_c_ may be a part of the first area **120**_a_ and/or the second area **130**_b_, and may be disposed at a location corresponding to the hinge device (e.g., hinge device **140** of FIG. **1**B). According to an embodiment, the electronic device **100** may include a hinge housing **141** (e.g., hinge cover) supporting the hinge device (e.g., hinge device **140** of FIG. **1**B). In an embodiment, the hinge housing **141** may be disposed to be exposed to outside when the electronic device **100** is in a folded state, and not to be viewed from the outside as being drawn into an inner space of the first housing **110** and an inner space of the second housing **120** when the electronic device **100** is in an unfolded state.

[0059] According to various embodiments, the electronic device **100** may include a second display **131** (e.g., sub-display) disposed separately from the first display **130**. According to an embodiment, the second display **131** may be disposed to be exposed at least partly on the second surface **112** of the first housing **110**. In an embodiment, when the electronic device **100** is in the folded state, the second display **131** may display at least a part of state information of the electronic device **100** in replacement of at least a part of a display function of the first display **130**. According to an embodiment, the second display **131** may be disposed to be viewed from the outside through at least a partial area of the first rear cover **114**. In an embodiment, the second display **13 1** may be disposed on the fourth surface **122** of the second housing **120**. In this case, the second display **131** may be disposed to be viewed from the outside through at least a partial area of the second rear cover **124**.

[0060] According to various embodiments, the electronic device **100** may include at least one of an input device **103** (e.g., microphone), sound output devices **101** and **102**, a sensor module **104**, camera devices **105** and **108**, a key input device **106**, or a connector port **107**. In an illustrated embodiment, although the input device **103** (e.g., microphone), the sound output devices **101** and **102**, the sensor module **104**, the camera devices **105** and **108**, the key input device **106**, or the connector port **107** are illustrated as hole or circular shaped elements formed on the first housing **110** or the second housing **120**, they are exemplarily illustrated for explanation, but are not limited thereto. According to various embodiments, the input device **103** may include at least one microphone **103** disposed on the second housing **120**. In an embodiment, the input device **103** may include a plurality of microphones **103** disposed to sense the sound direction. In an embodiment, the plurality of microphone **103** may be disposed at proper locations on the first housing **110** and/or the second housing **120**. According to an embodiment, the sound output devices **101** and **102** may include at least one speaker **101** and **102**. According to an embodiment, the at least one speaker **101** and **102** may include a receiver **101** for call disposed on the first housing **110**, and the speaker **102** disposed on the second housing **120**. In an embodiment, the input device **103**, the sound output devices **101** and **102**, and the connector port **107** may be disposed in a space provided in the first housing **110** and/or the second housing **120** of the electronic device **100**, and may be exposed to an external environment through at least one hole formed on the first housing **110** and/or the second housing **120**. According to an embodiment, the at least one connector port **107** may be used to transmit and receive power and/or data to and from an external electronic device. In an embodiment, the at least one connector port (e.g., ear jack hole) may accommodate a connector (e.g., ear jack) for transmitting and receiving an audio signal to and from the external electronic device. In an embodiment, the hole formed on the first housing **110** and/or the second housing **120** may be commonly used for the input device **103** and the sound output devices **101** and **102**. In an embodiment, the sound output devices **101** and **102** may include a speaker (e.g., piezo-electric speaker) that is not exposed through the hole formed on the first housing **110** and/or the second housing **120**.

[0061] According to various embodiments, the sensor module **104** may generate an electrical signal or a data value corresponding to an internal operation state of the electronic device **100** or an external environment state. According to an embodiment, the sensor module **104** may detect the external environment through the first surface **111** of the first housing **110**. In an embodiment, the electronic device **100** may further include at least one sensor module disposed to detect the external environment through the second surface **112** of the first housing **110**. According to an embodiment, the sensor module **104** (e.g., illumination sensor) may be disposed to detect the external environment through the first display **130** under the first display **130**. According to an embodiment, the sensor module **104** may include at least one of a gesture sensor, a gyro sensor, a barometric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a color sensor, an infrared (IR) sensor, a biosensor, a temperature sensor, a humidity sensor, an illumination sensor, a proximity sensor, a biosensor, an ultrasonic sensor, or an illumination sensor **104**.

[0062] According to various embodiments, the camera devices **105** and **108** may include the first camera device **105** (e.g., front camera device) disposed on the first surface **111** of the first housing **110**, and the second camera device **108** disposed on the second surface **112** of the first housing **110**. In an embodiment, the electronic device **100** may further include a flash **109** disposed near the second camera device **108**. According to an embodiment, the camera devices **105** and **108** may include at least one lens, an image sensor, and/or an image signal processor. According to an embodiment, the camera devices **105** and **108** may be disposed so that two or more lenses (e.g., wide angle lens, ultra wide angle lens, or telephoto lens) and two or more image sensors are located on one surface (e.g., first surface **111**, second surface **112**, third surface **121**, or fourth surface **122**) of the electronic device **100**. In an embodiment, the camera devices **105** and **108** may include lenses for time of flight (TOF) and/or image sensors.

[0063] According to various embodiments, the key input device **106** (e.g., key buttons) may be disposed on the third side surface **113**_c_ of the first lateral member **113** of the first housing **110**. In an embodiment, the key input device **106** may be disposed on at least one side surface of other side surfaces **113***a* and **113***b* of the first housing **110** and/or side surfaces **123***a*, **123***b*, and **123***c* of the second housing **120**. In an embodiment, the electronic device **100** may not include some or all of the key input devices **106**, and the non-included key input device **106** may be implemented in another form, such as a soft key, on the first display **130**. In an embodiment, the key input device **106** may be implemented by using the pressure sensor included in the first display **130**.

[0064] According to various embodiments, one of the camera devices **105** and **108** (e.g., first camera device **105**) or the sensor module **104** may be disposed to be exposed through the first display **130**. According to an embodiment, the first camera device **105** or the sensor module **104** may be optically exposed to the outside through an opening (e.g., through-hole) formed at least partly on the first display **130** in the inner space of the electronic device **100**. According to an embodiment, at least a part of the sensor module **104** may be disposed not to be visually exposed through the first display **130** in the inner space of the electronic device **100**. With reference to FIG. **2**B, the electronic device **100** may operate to maintain at least one designated folding angle in an intermediate state through the hinge device (e.g., hinge device **140** of FIG. **1**B). In this case, the electronic device **100** may control the first display **130** so that different kinds of content are displayed on the display area corresponding to the first surface **111** and the display area corresponding to the third surface **121**. According to an embodiment, the electronic device **100** may operate in substantially the unfolded state (e.g., unfolded state of FIG. **1**A) and/or in substantially the folded state (e.g., folded state of FIG. **2**A) based on a specific folding angle (e.g., angle between the first housing **110** and the second housing **120** when the electronic device **100** is in the intermediate state) through the hinge device (e.g., hinge device **140** of FIG. **1**B). In an embodiment, if a pressing force is provided in an unfolding direction (direction A) in a state where the electronic device **100** is unfolded at the specific folding angle through the hinge device (e.g., hinge device **140** of FIG. **1**B), the electronic device **100** may operate to be transitioned to the unfolded state (e.g., unfolded state of FIG. **1**A). In an embodiment, if the pressing force is provided in a folding direction (direction B) in a state where the electronic device **100** is unfolded at the specific folding angle through the hinge device (e.g., hinge device **140** of FIG. **1**B), the electronic device **100** may operate to be transitioned to the folded state (e.g., folded state of FIG. **2**A). In an embodiment, the electronic device **100** may operate to maintain the unfolded state (not illustrated) at various angles through the hinge device (e.g., hinge device **140** of FIG. **1**B) (free-stop function).

[0065] FIG. **3** is an exploded perspective view of the electronic device according to one or more embodiments of the present disclosure.

[0066] With reference to FIG. **3**, the electronic device **100** may include a first housing **110** (e.g., a first housing structure) including a first lateral member **113** (e.g., a first lateral frame or a first lateral bezel), a second housing **120** (e.g., a second housing structure) which is foldably connected to the first housing **110** through a hinge device **140** and includes a second lateral member **123** (e.g., a second lateral frame or a second lateral bezel), and a first display **130** (e.g., a flexible display) disposed to be supported by the first housing **110** and the second housing **120**. In an embodiment, the first housing **110** may include a first rear cover **114** coupled with the first lateral member **113**. In an embodiment, the first housing **110** may include a first support member **1131** (e.g., a first support plate) that extends at least partially from the first lateral member **113**, and includes a first surface **1131***a* and a second surface **1131***b*, which faces in the opposite direction to the first surface **1131***a*. In an embodiment, the first housing **110** may include a first space (e.g., first space **1101** in FIG. **4**) formed between the second surface **1131***b* of the first support member **1131** and the first rear cover **114**. In an embodiment, the second housing **120** may include a second rear cover **124** coupled with the second lateral member **123**. In an embodiment, the second housing **120** may include a second support member **1231** (e.g., a second support plate) that extends at least partially from the second lateral member **123**, and includes a third surface **1231***a*, which faces in the same direction as the first surface **1131***a*, and a fourth surface **1231***b*, which faces in the opposite direction to the third surface **1231***a*, in the unfolding state. In an embodiment, the second housing

120 may include a second space (e.g., second space **1201** in FIG. **4**) formed between the fourth surface **1231***b* and the second rear cover **124**. In an embodiment, the first support member **1131** may be integrally formed with the first lateral member **113** or may be structurally coupled with the first lateral member **113**. In an embodiment, the second support member **1231** may be integrally formed with the second lateral member **123** or structurally coupled with the second lateral member **123**. In an embodiment, the first display **130** may be disposed to be supported by the first surface **1131***a* of the first support member **1131** and the third surface **1231***a* of the second support member **1231**. In an embodiment, the first lateral member **113** and the first rear cover **114** may be integrally formed. In an embodiment, the second lateral member **123** and the second rear cover **124** may be integrally formed. In an embodiment, the electronic device **100** may include the second display **131** disposed in the first space **1101** and externally visible through at least some area of the first rear cover **114**.

[0067] According to one or more embodiments, the electronic device **100** may include a first printed circuit board **151** (e.g., first printed circuit board assembly or main printed circuit board) disposed in the first space **1101** and at least one electronic component (e.g., camera module **105**, receiver **101** or first batteries B**1**) disposed on the first printed circuit board **151**. In an embodiment, the electronic device **100** may include a second printed circuit board **152** (e.g., second printed circuit board assembly or sub printed circuit board) disposed in the second space **1201**, and at least one electronic component (e.g., speaker **102** or batteries B**2**) disposed on the second printed circuit board. In an embodiment, the electronic device may include a flexible printed circuit board **160** (e.g., flexible substrate, FPCB, or wiring member) that electrically connects the first printed circuit board **151** and the second printed circuit board **152** through the hinge device **140**. In an embodiment, the electronic device **100** may include a hinge housing **141** disposed in an area corresponding to the hinge device **140**. In an embodiment, the hinge device **140** may be disposed by the hinge housing **141** to not be visible from the outside of the electronic device **100**. In an embodiment, at least a portion of the flexible printed circuit board **160** may be disposed in a space (e.g., the space **1411** in FIG. **4**) formed by the hinge housing **141** to be elastically deformable and not visible from the outside.

[0068] According to an example embodiment of the present disclosure, the flexible printed circuit board **160** may be disposed in an area corresponding to the hinge device **140** (e.g., the space **1411** in FIG. **4**) to be bent once or more in different directions. In an embodiment, the bent area of the flexible printed circuit board **160** may accommodate the length changes of the flexible printed circuit board **160**, which may stretch or shrink according to the folding operation of the electronic device **100**. In an embodiment, the electronic device **100** may include a waterproof structure **200** and **200-1** for the flexible printed circuit board **160**. For example, the waterproof structure **200** and **200-1** may include a first waterproof structure **200** that seals a first through-hole (e.g., the first through-hole **1131***c* in FIG. **4**) formed in the first support member **1131** to block external moisture or foreign substances when the flexible printed circuit board **160** is electrically connected to the first printed circuit board **151** through the first through-hole **1131***c*, and a second waterproof structure **200-1** that seals a second through-hole **1231***c* formed in the second support member **1231** to block external moisture or foreign substances when the flexible printed circuit board **160** is electrically connected to the second printed circuit board **152** through the second through-hole **1231***c*. In an embodiment, the electronic device **100** may include a fixation structure configured to prevent the movement of the flexible printed circuit board **160**, which is accommodated in the first housing **110** and second housing **120**, when the electronic device **100** performs a folding operation. The electronic device **100** according to an example embodiment of the present disclosure, in which the fixation structure is configured together with the waterproof structure **200** and **200-1**, may help in slimming down the electronic device **100** and/or improving assembly by securing a sufficient bending area for the flexible printed circuit board **160** (e.g., by securing enough driving length for the flexible printed circuit board), without increasing the thickness of the electronic device **100**,

compared to a layout structure where the fixation structure is disposed separately.
[0069] FIG. **4** is a partial cross-sectional view of the electronic device taken along line **4**-**4** of FIG. **1**C according to one or more embodiments of the present disclosure.
[0070] With reference to FIG. **4**, the electronic device **100** may include the second housing **120** that is rotatably coupled relative to the first housing **110** through a hinge device (e.g., the hinge device **140** in FIG. **3**), and a flexible display **130** (e.g., the first display **130** in FIG. **3**) disposed to be supported by the first housing **110** and the second housing **120**. In an embodiment, the first housing **110** may include the first lateral member **113** and the first rear cover **114**, which is coupled with the first lateral member **113** and forms the first space **1101**. In an embodiment, the second housing **120** may include the second lateral member **123** and the second rear cover **124**, which is coupled with the second lateral member **123** and forms the second space **1201**. In an embodiment, the first lateral member **113** may at least partially form the appearance of the side surface (e.g., the first side surface **113***a*, the second side surface **113***b*, and the third side surface **113***c* in FIG. **1**A) of the electronic device **100**, and may include the first support member **1131** that extends from the side surface into at least a portion of the first space **1101**. In an embodiment, the first space **1101** may be provided through the coupling of the first support member **1131** and the first rear cover **114**. In an embodiment, the second lateral member **123** may at least partially form the appearance of the side surface (e.g., the fourth side surface **123***a*, the fifth side surface **123***b*, and the sixth side surface **123***c* in FIG. **1**A) of the electronic device **100**, and may include the second support member **1231** that extends from the side surface into at least a portion of the second space **1201**. In an embodiment, the second space **1201** may be provided through the coupling of the second support member **1231** and the second rear cover **124**. In an embodiment, the first support member **1131** may include the first surface **1131***a* facing a first direction (e.g., the z-axis direction) and the second surface **1131***b* facing a second direction (e.g., the −z-axis direction), opposite to the first direction, in the unfolding state. In an embodiment, the second support member **1231** may include the third surface **123** la facing the same direction as the first surface **1131***a*, and the fourth surface **1231***b* facing the opposite direction to the third surface **1231***a*, in the unfolding state. In an embodiment, the flexible display **130** may be disposed to be supported by the first surface **1131***a* and the third surface **1231***a*. In an embodiment, the electronic device **100** may include the hinge housing **141** that includes the space (an accommodation space) **1411** that is disposed between the first housing **110** and the second housing **120** and is intended to accommodate the hinge device (e.g., the hinge device **140** in FIG. **3**), such that the hinge device is not visible from the outside. In an embodiment, the hinge housing **141** may be disposed in such a manner that, when the electronic device **100** is in the folding state, it is at least partially visible from the outside, and in the unfolding state, it is concealed by the first housing **110** and second housing **120** to be not visible from the outside.
[0071] According to one or more embodiments, the electronic device **100** may include at least one flexible printed circuit board **160** (e.g., the FPCB, the flexible printed circuit board, or a flat ribbon cable (FRC), a flexible RF cable) to electrically connect at least one first electronic component disposed in the first space **1101** of the first housing **110** and at least one second electronic component disposed in the second space **1201** of the second housing **120**. In an embodiment, at least one first electronic component may include a main printed circuit board (e.g., the first printed circuit board **151** in FIG. **3**). In an embodiment, at least one second electronic component may include a sub printed circuit board (e.g., the second printed circuit board **152** in FIG. **3**). In an embodiment, at least one first electronic component and/or at least one second electronic component may include at least one battery B**1** or B**2** disposed in the first space **1101** and/or second space **1201**, at least one antenna radiator, at least one sensor module, or at least one input/output device. For example, the flexible printed circuit board **160** may be disposed to electrically connect various electronic components disposed in the first space **1101** and the second space **1201**. In an embodiment, the flexible printed circuit board **160** may be disposed in such a manner that it penetrates the first through-hole **1131***c* formed in the first support member **1131** of the first housing

110 from the first space **1101**, passes through the space (the accommodation space) **1411** of the hinge housing **141**, and after penetrating the second through-hole **1231***c* formed in the second support member **1231** of the second housing **120**, it is accommodated into the second space **1201**. In an embodiment, the flexible printed circuit board **160** may be disposed in the space (the accommodation space) **1411** of the hinge housing **141** to have a shape and extra length for accommodating length changes due to the folding operation of the electronic device **100**. For example, the flexible printed circuit board **160** may be formed to have a shape in the space (the accommodation space) **1411** that is bent at least once and capable of being elastically restored.
[0072] According to one or more embodiments, the electronic device **100** may include the first waterproof structure **200** to seal the first through-hole **1131***c* and the second waterproof structure **200-1** to seal the second through-hole **1231***c*. In an embodiment, the first waterproof structure **200** and the second waterproof structure **200-1** may prevent or reduce the infiltration of external moisture and/or foreign substances into the first space **1101** and the second space **1201** through the first through-hole **1131***c* and the second through-hole **1231***c*. In an embodiment, the first waterproof structure **200** may include a first waterproof member **210** that is coupled with a stiffener **161** fixed to the first support member **1131** and disposed on the flexible printed circuit board **160**. In an embodiment, the first waterproof member **210** may include a loop structure (e.g., a closed loop structure) having an opening (e.g., the opening **214** in FIG. **6**B) through which the flexible printed circuit board **160** passes, by being coupled with the stiffener **161**. In an embodiment, the first waterproof member **210** and the stiffener **161** may be disposed on the first support member **1131** in such a manner that the loop structure (or the opening **214** in FIG. **6**B) surrounds the periphery of the first through-hole **1131***c*. In an embodiment, the first waterproof structure **200** may include a second waterproof member **220** disposed to seal the opening **214**, and a fixing bracket **230** that is coupled with the first support member **1131** above the second waterproof member **220**. In an embodiment, the first waterproof structure **200** may include the first support member **1131** and a third waterproof member **240** disposed between the first waterproof member **210** and stiffener **161**. In an embodiment, the second waterproof structure **200-1** may have a configuration that is substantially identical to the first waterproof structure **200** and may seal the second through-hole **1231***c*.
[0073] The electronic device **100** according to an example embodiment of the present disclosure may have a fixation structure for the stiffener **161**, which is used to fix the flexible printed circuit board **160**, configured together with the waterproof structure **200** and **200-1**. This configuration helps secure a sufficient bending area **1142** for the flexible printed circuit board **160** according to the folding operation without increasing the thickness of the electronic device **100**, thereby assisting in slimming down the electronic device **100** and/or improving assembly.
[0074] Hereinafter, the first waterproof structure **200** disposed on the first support member **1131** of the first housing **110** is illustrated and described. However, the second waterproof structure **200-1** disposed on the second support member **1231** of the second housing **120** may also have a layout configuration that is substantially identical.
[0075] FIG. **5**A is an exploded perspective view of a waterproof structure according to one or more embodiments of the present disclosure. FIG. **5**B illustrates the assembly sequence of the waterproof structure according to one or more embodiments of the present disclosure.
[0076] FIGS. **6**A to **6**G illustrate the coupling process according to one or more embodiments of the present disclosure.
[0077] With reference to FIGS. **5**A to **6**G, the electronic device **100** may include the first housing **110** including the first lateral member **113**. In an embodiment, the first housing **110** may extend from the first lateral member **113** and include the first surface **1131***a* facing a first direction (e.g., the z-axis direction) and the second surface **1131***b* facing a direction opposite to the first surface **1131***a* (e.g., the −z-axis direction). In an embodiment, the first lateral member **113** may include a through-hole **1131***c* that penetrates from the first surface **1131***a* to the second surface **1131***b*. In an

embodiment, the through-hole **1131***c* may be formed in a recess **1131***d* formed on the second surface **1131***b* of the first support member **1131**. In an embodiment, the recess **1131***d* may be formed lower than the second surface **1131***b*. In an embodiment, the through-hole **1131***c* may be formed to have a length in a third direction (e.g., the ±x-axis direction) that is perpendicular to the first direction and parallel to the surface of the flexible display **130**.

[0078] According to one or more embodiments, the electronic device **100** may include the flexible printed circuit board **160** that passes through the through-hole **1131***c*. In an embodiment, the flexible printed circuit board **160** may penetrate through the through-hole **1131***c* from the first space (e.g., the first space **1101** in FIG. **4**) of the first housing **110** and then be withdrawn into the accommodate space (e.g., the space **1411** in FIG. **4**) of the hinge housing (e.g., the hinge housing **141** in FIG. **4**). In an embodiment, the flexible printed circuit board **160** may include the stiffener **161** that is at least partially attached or formed and disposed near the through-hole **1131***c* in the recess **1131***d*. In an embodiment, the stiffener **161** may be coupled to the flexible printed circuit board **160** and may be formed of a metal or polymer material. In an embodiment, the both ends of the stiffener **161** may include a pair of coupling grooves **1611** and **1612** that are recessed from the periphery.

[0079] According to one or more embodiments, the electronic device **100** may include the waterproof structure **200** (e.g., the first waterproof structure **200** in FIG. **4**) disposed through the through-hole **1131***c* in the recess **1131***d* to seal the through-hole **1131***c* that the flexible printed circuit board **160** passes through. In an embodiment, the waterproof structure **200** may include the first waterproof member **210** that is at least partially coupled with the stiffener **161**, the second waterproof member **220** that is coupled with the first waterproof member **210** and the stiffener **161**, and the fixing bracket **230** disposed above the second waterproof member **220** to fix the first waterproof member **210**, the second waterproof member **220** and the stiffener **161** within the recess **1131***d*. In an embodiment, the fixing bracket **230** may include a plate portion **231** that is disposed to at least partially overlap the second waterproof member **220**, and a pair of extensions **232** and **233** that extend from both ends of the plate portion **231** and are designed to be fastened into screw fastening holes **1135** formed in the recess **1131***d* using screws S. In an embodiment, the waterproof structure **200** may include the third waterproof member **240** applied to the recess **1131***d* along the periphery of the through-hole **1131***c* before the stiffener **161** and the first waterproof member **210** are disposed in the recess **1131***d*. In some embodiments, the third waterproof member **240** may be omitted.

[0080] According to one or more embodiments, the first waterproof member **210** may include a body **211** and a pair of coupling protrusions **212** and **213** formed at both ends of the body **211**. In an embodiment, the first waterproof member **210** may have a partially opened loop shape, for example. In an embodiment, the first waterproof member **210** may be one opened side. In an embodiment, the body **211** may be formed of a rigid material, and the pair of coupling protrusions **212** and **213** may be formed of a soft material with elasticity, which is coupled with the body **211**. In an embodiment, the body **211** may be formed of metal or polycarbonate (PC). In an embodiment, the pair of coupling protrusions **212** and **213** may be formed of an elastic material such as silicone, rubber, or urethane. In an embodiment, the body **211** and the pair of coupling protrusions **212** and **213** may be coupled through injection molding. In some embodiments, the body **211** and the pair of coupling protrusions **212** and **213** may be coupled through an attachment manner such as bonding or taping. In some embodiments, the body **211** and the pair of coupling protrusions **212** and **213** may be coupled through a structural shape. In an embodiment, the stiffener **161** may include the pair of coupling protrusions **212** and **213**. For example, the pair of coupling protrusions **212** and **213** may be formed of a soft material with elasticity, which is coupled with the stiffener **161**. For example, the stiffener **161** and the pair of coupling protrusions **212** and **213** may be coupled through an attachment manner such as bonding or taping. In some embodiments, the stiffener **161** and the pair of coupling protrusions **212** and **213** may be coupled through a structural shape. In some embodiments, the coupling protrusions **212** and **213** may be formed on the stiffener. In this case, the coupling grooves **1611** and **1612**, which are to couple with the coupling protrusions **212** and **213**, may be formed on the first waterproof member **210**. For example, the coupling grooves that couple with the body **211** may also be formed of an elastic material.

[0081] Hereinafter, the assembly sequence of the waterproof structure **200** is as follows.

[0082] With reference to FIGS. **5**B to **6**G, in the first operation **501**, before the waterproof structure **200** is applied to the electronic device **100**, the stiffener **161** of the flexible printed circuit board **160** and the first waterproof member **210** may be coupled first. In an embodiment, the first waterproof member **210** may be coupled in a manner where the pair of coupling protrusions **212** and **213** are inserted (e.g., press-fit) into the pair of coupling grooves **1611** and **1612** of the stiffener **161**. In an embodiment, the pair of coupling protrusions **212** and **213** are formed of a soft material with elasticity, and are therefore tightly coupled to overlap the pair of coupling grooves **1611** and **1612**, which helps maintain the waterproof functionality. In an embodiment, the opened loop shape of the first waterproof member **210** may be transformed into a closed loop shape through coupling with the stiffener **161**, thereby forming the opening **214**. In other words, an opened side of the first waterproof member **210** may be closed by a coupling with the stiffener **161**, thus the first waterproof member **210** and the stiffener **161**, in combination, may form the opening **214**. Therefore, the flexible printed circuit board **160** may be penetrated through the through-hole **1131***c* below the opening **214**.

[0083] Then, as illustrated in FIG. **6**C, the third waterproof member **240** may be disposed in the recess **1131***d* to surround the periphery of the through-hole **1131***c*. In an embodiment, the third waterproof member **240** may include a waterproof tape (e.g., a double-sided tape) attached to the recess **1131***d*. In an embodiment, the third waterproof member **240** may be attached over the entire area of the recess **1131***d*, except for the through-hole **1131***c* and screw fastening holes **1135**. In some embodiments, the third waterproof member **240** may be omitted.

[0084] In the second operation **503**, as illustrated in FIG. **6**D, the stiffener **161** and the first waterproof member **210** coupled with the stiffener **161** may be seated in the recess **1131***d*. In this case, the stiffener **161** and the first waterproof member **210** coupled with the stiffener **161** may be disposed above the third waterproof member **240**. In an embodiment, the closed loop structure (or the opening) formed by the stiffener **161** and the first waterproof member **210** may be disposed to contact along the periphery of the through-hole **1131***c*. For example, when the first waterproof member **210** is viewed from above, or when the second surface **1131***b* is viewed from above, the opening **214** may overlap the through-hole **1131***c*. In an embodiment, the opening **214** is at least partially aligned with the through-hole **1131***c*. In an embodiment, the opening **214** may be smaller than the through-hole **1131***c*. In some embodiments, the opening **214** may be formed to be substantially the same size as the through-hole **1131***c*. In some embodiments, the opening **214** may be formed larger than the through-hole **1131***c*. In an embodiment, once the stiffener **161** and the first waterproof member **210** coupled with the stiffener **161** are seated in the recess **1131***d*, the flexible printed circuit board **160** may be penetrated through the through-hole **1131***c*.

[0085] In the third operation **505**, as illustrated in FIG. **6**E, the second waterproof member **220** may be disposed above the stiffener **161** and the first waterproof member **210** to cover the opening **214**. In an embodiment, the second waterproof member **220** may include a waterproof tape. In an embodiment, the second waterproof member **220** may include double-sided tape.

[0086] In the fourth operation **507**, as illustrated in FIGS. **6**F and **6**G, the fixing bracket **230** may be fixed in such a manner that the plate portion **231** is disposed above the second waterproof member **220**, and the extensions **232** and **233** at both ends are securely fastened into the screw fastening holes **1135** formed in the recess **1131***d* using screws S. In an embodiment, the plate portion **231** of the fixing bracket **230** may be formed to cover at least the opening **214** when the first waterproof member **210** is viewed from above, and may be disposed to overlap the opening **214**. Therefore, the opening **214** covering the through-hole **1131***c* may be sealed through the second waterproof member **220** and the third waterproof member **240**. In addition, the first waterproof member **210**, the second waterproof member **220**, and the third waterproof member **240**, along with the stiffener **161**, may be securely fixed in the recess **1131***d* of the first housing **110** through the fixing bracket **230**. In an embodiment, after the waterproof structure **200** is fully assembled, the plate portion **231** of the fixing bracket **230** may be configured to align with the second surface **1131***b* of the first support member **1131** or be disposed lower than the second surface **1131***b* within the recess **1131***d*.

[0087] The electronic device **100** according to an example embodiment of the present disclosure, where the fixation structure of the stiffener **161** for fixing the flexible printed circuit board **160** is configured together with the waterproof structure **200**, may help in slimming down the electronic device **100** and/or improving assembly by securing a sufficient bending area for the flexible printed circuit board **160** due to the folding operation, without increasing the thickness of the electronic device **100**.

[0088] According to one or more embodiments, an electronic device may include a first housing (e.g., the first housing **110** in FIG. **3**) including at least one first electronic component (e.g., the first printed circuit board **151** in FIG. **3**), a second housing (e.g., the second housing **120** in FIG. **3**) including at least one second electronic component (e.g., the second printed circuit board **152** in FIG. **3**), a hinge device (e.g., the hinge device **140** in FIG. **3**) that rotatably connects the first housing and the second housing, a flexible printed circuit board (e.g., the flexible printed circuit board **160** in FIG. **4**) that penetrates through a through-hole (e.g., the through-holes **1131***c* and **1231***c* in FIG. **4**) formed in the first housing and/or the second housing, and electrically connects at least one first electronic component and at least one second electronic component, and a waterproof structure (e.g., the first waterproof structure **200** and the second waterproof structure **200-1** in FIG. **4**) disposed in the through-hole, in which the waterproof structure (e.g., the first waterproof structure **200** in FIG. **4**) may include a first waterproof member (e.g., the first waterproof member **210** in FIG. **5**A) coupled with a stiffener (e.g., the stiffener **161** in FIG. **4**) disposed on the flexible printed circuit board to form a loop structure that includes an opening (e.g., the opening **214** in FIG. **6**B), and a second waterproof member (e.g., the second waterproof member **220** in FIG. **5**A) disposed on the stiffener and the first waterproof member to cover the opening, in which the stiffener and the first waterproof member may be disposed on the first housing and/or the second housing in such a manner that the loop structure (or the opening) surrounds the periphery of the through-hole.

[0089] According to one or more embodiments, the first waterproof member has an opened side that is closed by a coupling with the stiffener.

[0090] According to one or more embodiments, when the through-hole is viewed from above, the opening may overlap the through-hole.

[0091] According to one or more embodiments, the opening may be formed smaller than the through-hole.

[0092] According to one or more embodiments, the first waterproof member may include a body formed of a rigid material, and a pair of coupling protrusions formed of a soft material with elasticity, each coupled to both ends of the body.

[0093] According to one or more embodiments, the stiffener may include a pair of coupling grooves formed at both ends to couple with the pair of coupling protrusions.

[0094] According to one or more embodiments, the body may be formed of metal, and the pair of coupling protrusions may be formed of at least one of rubber, urethane, or silicone.

[0095] According to one or more embodiments, the electronic device may include a third waterproof member disposed between the first housing and/or second housing, and the stiffener and the first waterproof member. The third waterproof member may be disposed to cover the through-hole and/or the opening.

[0096] According to one or more embodiments, the second waterproof member and third waterproof member may include a waterproof tape.

[0097] According to one or more embodiments, the electronic device may include a fixing bracket that is disposed above the second waterproof member, overlapping at least a portion of the second waterproof member, and fixed to the first housing and/or the second housing.

[0098] According to one or more embodiments, the fixing bracket may include a plate portion disposed to overlap the second waterproof member, and a pair of extensions extending from both ends of the plate portion, which are configured to be fixed to the first housing and/or the second housing.

[0099] According to one or more embodiments, the fixing bracket may be fixed in such a manner that the pair of extensions are fastened to the first housing and/or the second housing through screws.

[0100] According to one or more embodiments, the through-hole may be formed in a recess formed in the first housing and/or the second housing.

[0101] According to one or more embodiments, the waterproof structure may be disposed not to protrude from the recess.

[0102] According to one or more embodiments, the at least one first electronic component may include a main printed circuit board, and the at least one second electronic component may include a sub printed circuit board.

[0103] According to one or more embodiments, an electronic device may include a first housing (e.g., the first housing **110** in FIG. **3**) including at least one first electronic component (e.g., the first printed circuit board **151** in FIG. **3**), a second housing (e.g., the second housing **120** in FIG. **3**) including at least one second electronic component (e.g., the second printed circuit board **152** in FIG. **3**), a hinge device (e.g., the hinge device **140** in FIG. **3**) that rotatably connects the first housing and the second housing, a flexible printed circuit board (e.g., the flexible printed circuit board **160** in FIG. **4**) that penetrates through through-holes (e.g., the through-holes **1131***c* and **1231***c* in FIG. **4**) formed each in the first housing and the second housing, and electrically connects at least one first electronic component and at least one second electronic component, and waterproof structures (e.g., the first waterproof structure **200** and the second waterproof structure **200-1** in FIG. **4**) each disposed in the through-holes, in which the waterproof structure (e.g., the first waterproof structure **200** in FIG. **4**) may include a first waterproof member (e.g., the first waterproof member **210** in FIG. **5**A) coupled with a stiffener (e.g., the stiffener **161** in FIG. **4**) disposed on the flexible printed circuit board to form a loop structure that includes an opening (e.g., the opening **214** in FIG. **6**B), a second waterproof member (e.g., the second waterproof member **220** in FIG. **5**A) disposed on the stiffener and the first waterproof member to cover the opening, a third waterproof member (e.g., the third waterproof member **240** in FIG. **5**A) disposed between the first housing and/or second housing, and the stiffener and the first waterproof member, and a fixing bracket (e.g., the fixing bracket **230** in FIG. **5**A) disposed to overlap at least part of the second waterproof member and fixed to the first housing and/or the second housing, in which the stiffener and the first waterproof member may be disposed on the first housing and the second housing in such a manner that the loop structure (or the opening) surrounds the periphery of the through-hole.

[0104] According to one or more embodiments, the first waterproof member has an opened side that is closed by a coupling with the stiffener.

[0105] According to one or more embodiments, when the through-hole is viewed from above, the opening may overlap the through-hole.

[0106] According to one or more embodiments, the first waterproof member may include a body formed of a rigid material, and a pair of coupling protrusions formed of a soft material with elasticity, each coupled to both ends of the body.

[0107] According to one or more embodiments, the stiffener may include a pair of coupling grooves formed at both ends to couple with the pair of coupling protrusions.

[0108] Further, the embodiments of the present disclosure disclosed in the present specification and illustrated in the drawings are provided as particular examples for easily explaining the technical contents according to the embodiment of the present disclosure and helping understand the embodiment of the present disclosure, but not intended to limit the scope of the embodiment of the present disclosure. Accordingly, the scope of the one or more embodiments of the present disclosure may be interpreted as including all alterations or modifications derived from the technical spirit of the one or more embodiments of the present disclosure in addition to the disclosed embodiments.

Claims

1. An electronic device comprising: a first housing comprising at least one first electronic component, the first housing having a through-hole therein; a second housing comprising at least one second electronic component; a hinge device that rotatably connects the first housing to the second housing; a flexible printed circuit board that penetrates through the through-hole in the first housing, and electrically connects the at least one first electronic component and the at least one second electronic component of the second housing; a stiffener on the flexible printed circuit board; and a waterproof structure in the through-hole, wherein the waterproof structure comprises: a first waterproof member coupled with the stiffener to form a loop structure having an opening, and a second waterproof member on the first waterproof member and covering the opening, and wherein the stiffener and the first waterproof member surround a periphery of the through-hole.
2. The electronic device of claim 1, wherein the first waterproof member has an opened side that is closed by coupling with the stiffener.
3. The electronic device of claim 1, wherein the opening is at least partially aligned with the through-hole.
4. The electronic device of claim 1, wherein the opening is smaller than the through-hole.
5. The electronic device of claim 1, wherein the first waterproof member comprises: a body made of a rigid material; and a first coupling protrusion and a second coupling protrusion respectively coupled to opposing sides of the body and each made of an elastic material.
6. The electronic device of claim 5, wherein the stiffener comprises a first coupling groove and a second coupling groove respectively at opposite ends of the stiffener, the first coupling groove being coupled to the first coupling protrusion and the second coupling groove being coupled to the second coupling protrusion.
7. The electronic device of claim 5, wherein the rigid material of the body comprises a metal, and the elastic material of the first and second coupling protrusions comprises at least one of rubber, urethane, or silicone.
8. The electronic device of claim 1, further comprising a third waterproof member disposed between the stiffener and the first housing and between the first waterproof member and the first housing, wherein the third waterproof member surrounds the through-hole or the opening.
9. The electronic device of claim 8, wherein at least one of the second waterproof member or the third waterproof member comprises a waterproof tape.
10. The electronic device of claim 1, further comprising a fixing bracket above the second waterproof member, overlapping at least a portion of the second waterproof member, and fixed to the first housing.
11. The electronic device of claim 10, wherein the fixing bracket comprises: a plate portion that overlaps the second waterproof member, and a pair of extensions extending from opposite ends of the plate portion, and wherein the pair of extensions are fixed to the first housing.
12. The electronic device of claim 11, wherein the fixing bracket is fixed in such a manner that the pair of extensions are fastened to the first housing through screws.
13. The electronic device of claim 1, wherein the through-hole is in a recess in the first housing.
14. The electronic device of claim 13, wherein the waterproof structure does not protrude from the

recess.

15. The electronic device of claim 1, wherein the at least one first electronic component comprises a main printed circuit board, and wherein the at least one second electronic component comprises a sub printed circuit board.

16. An electronic device comprises: a first housing comprising at least one first electronic component, the first housing having a first through-hole therein; a second housing comprising at least one second electronic component, the second housing having a second through-hole therein; a hinge device that rotatably connects the first housing to the second housing, a flexible printed circuit board that penetrates through the first through-hole in the first housing and the second through-hole in the second housing, and electrically connects the at least one first electronic component and the at least one second electronic component; a stiffener on the flexible printed circuit board; a first waterproof structure in the first through-hole; and a second waterproof structure in the second through-hole, wherein each of the first waterproof structure and the second waterproof structure comprises: a first waterproof member coupled with the stiffener to form a loop structure having an opening, a second waterproof member on the first waterproof member and covering the opening, a third waterproof member between the stiffener and the first waterproof member, and a fixing bracket overlapping at least a portion of the second waterproof member and fixed to a corresponding one of the first housing or the second housing, wherein the stiffener and the first waterproof member surrounds a first periphery of the first through-hole and a second periphery of the second through-hole.

17. The electronic device of claim 16, wherein the first waterproof member has an opened side that is closed by coupling with the stiffener.

18. The electronic device of claim 16, wherein, the opening overlaps at least one of the first through-hole or the second through-hole.

19. The electronic device of claim 16, wherein the first waterproof member comprises: a body comprising a rigid material, and a first coupling protrusion and a second coupling protrusion respectively coupled to opposing sides of the body and each comprising an elastic material.

20. The electronic device of claim 19, wherein the stiffener comprises a first coupling groove and a second coupling groove respectively at opposite ends of the stiffener, the first coupling groove being coupled to the first coupling protrusion and the second coupling groove being coupled to the second coupling protrusion.